

1) Encrypt the message "ENJOY YOUR JOURNEY"
using rail fence with depth 3 & 2

⇒ Depth 2

E J Y O R J O R N E Y
N O Y U J U N Y

Cipher:- E J Y O R O R E N O Y U J U N Y

Depth 3:-

E Y R J O Y U J U N Y
N O Y U J U N Y
J O O E

Cipher:- E Y R J O Y U J U N Y J O O E

2) Encrypt the message as per your choice using Rail
Fence with depth 2 & 3 PT is GOOD DAY

Depth 2

G O D A Y
O D A Y

Cipher:- G O D Y O D A

Depth 3

G D
O D A Y
O

G D O D A Y

3. Decrypt the Cipher "EDCSN40WLEJYREASONS",

Rail Fence depth 4

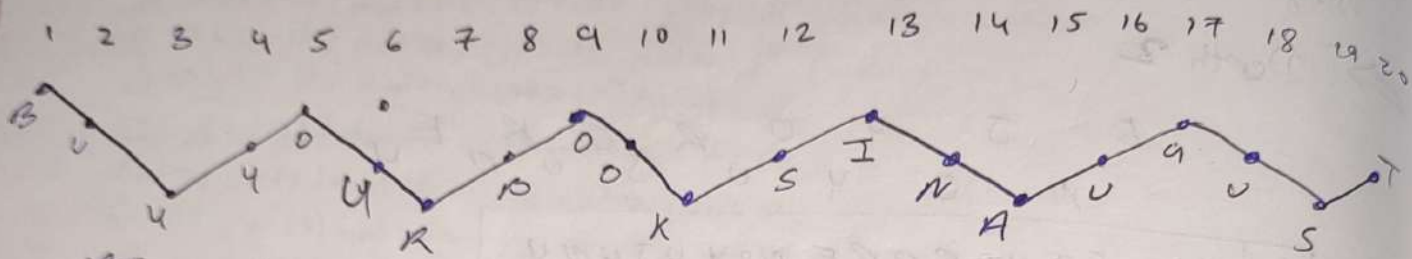
Depth = 4 length = 19

1 2 3 4 5 6 7 8 9 10

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19

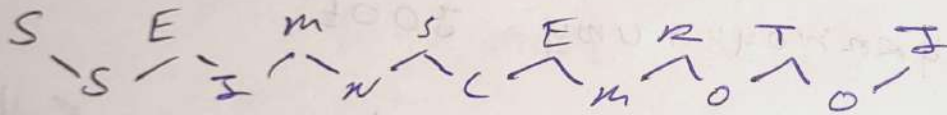
E N J O Y U J U N Y R E A S O N S
PT:- ENJOY YOUR NEW CLASSES

4) Decrypt the cipher "BOOKUYURBOSWUUTYRKAS",
 encrypted using Rail fence with depth 3
 dept = 3 length = 20



PT:- BUY YOUR BOOKS IN AUGUST

5) Decrypt the cipher "SEMSERTI SINCMOO" encrypted
 using rail fence with depth 2
 Dept 2 length 16



PT:- SSEIMNSCEMZOJ

Permutation:-

1. Encrypt the following message "we are discovered" using
 the Key 53124

5	3	1	2	4
W	E	A	R	E
D	I	S	C	O
V	E	R	E	D

CT:- ASRRCEEIEEODWDY

Permutation
1. Encrypt the following "We are discovered" using the key 53

DES:-

1. Using the initial permutation table, determine the output of the initial permutation box when the input is given in hexadecimal as: 0X002 000 0000 0001

Binary form

0000	0000	0000	0010
0000	0000	0000	0000
0000	0000	0000	0000
0000	0000	0000	0001

input box (PT)

0000 0000 0000 0010 0000 0000 0000 0000
0000 0000 0000 00000000 0010

Input bit 64=1, o/p position 25=1 } By using P-table
Input bit 15=1, o/p position 63=1

0000	0000	0000	0000	0000	0000	1000	0000	0000	0000
0000	0000	0000	0000	0010					

O/p hexadecimal

0X00000800000002

- 2) Given the 4 bit PT [P] 1101 & Key [K] 1011
calculate the resulting CT [C] using a bitwise Exclusive - OR [XOR] operation

Soln:- PT = 1101
Key = 1011
CT = 0110

XOR
Same bit = 0
diff bit = 1

∴ CT = 0110

- 3) The input to S box 1 is 000200 what is o/p

000000
↑ ↑
C C

0th row & 0th column in the S-box is 14

O/p = 1110

14 →

2	2	2	2
1	1	1	0